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Emerging Technologies Hackathon 2026

**PROJECT SUBMISSION TEMPLATE**

MACHINE HAWK

# Edge AI Predictive Maintenance Platform for Industrial Equipment

Real-Time Industrial Health Monitoring using Edge AI, ESP32 Microcontrollers, and Low-Latency MQTT Protocol.

`Edge AI Engine` `MQTT Protocol` `Predictive IIoT`


PROJECT NAME
Machine Hawk

TEAM NAME
Team Machine Hawk

TECHNOLOGY VERTICAL
**Semiconductors & Electronics / Edge AI**

# About the Team

TEAM NAME

Team Machine Hawk

INSTITUTE / ORGANISATION

Dr H N National College of Engineering

### Lead Systems Engineer

Hardware integration, ESP32 circuit design, MPU6050 & DHT22 sensor pinout calibration.

### Edge AI Engineer

Edge Impulse feature extraction, neural model training & C++ Arduino library deployment.

### Backend & Cloud Architect

Mosquitto MQTT broker configuration, Python analytics ingestion pipeline & database design.

### Frontend Web Developer

React, TypeScript & Tailwind CSS dashboard implementation with real-time UI sockets.

# Problem Statement & Industrial Challenge



### Unplanned Machine Outages

Catastrophic mechanical failures in rotating machinery (motors, pumps, gearboxes) halt entire production lines without prior warning.

### Traditional Maintenance Bottleneck

Calendar-based maintenance either services healthy machines too early or intervenes after catastrophic internal wear has already begun.

### Severe Downtime Costs

Idle manufacturing leads to substantial financial loss, ruined production batches, missed delivery schedules, and wasted labor hours.

### High-Cost Imported Solutions

Existing vibration monitoring tools are expensive imported systems requiring high-bandwidth cloud connectivity and ongoing subscription lock-ins.

### Reactive Maintenance Premium

Reactive fixes incur significantly higher repair expenses due to emergency part sourcing and severe secondary component damage.

### Core Objective

Addresses open innovation challenges for an indigenous, low-cost, low-power industrial predictive maintenance platform.

# Solution Overview & Innovation Novelty

Sensors › ESP32 › Edge AI › MQTT › Dashboard › Recommendation

### Local Edge AI Execution

Classifies vibration patterns on-device without continuous cloud dependency.

### Multi-Modal Telemetry

Correlates 3-axis acceleration changes with real-time temperature and humidity spikes.

### Zero-Latency MQTT

Ultra-lightweight publish/subscribe model ensuring instant alerting for plant engineers.

### Indigenous Alternative

Replaces expensive imported diagnostic hardware with locally producible micro-controller nodes.

# System Architecture & Technical Design

ARCHITECTURE & DESIGN

**Sensors**
MPU6050 & DHT22
RAW SAMPLING

→

**ESP32 Node**
Edge Impulse C++
TINYML INFERENCE

→

**MQTT Broker**
Mosquitto MQTTS
PUB/SUB TOPICS

→

**Python Backend**
FastAPI & DB
ANALYTICS ENGINE

### React & Tailwind Operator Interface

Renders live vibration metrics, Health Score gauges, failure risk levels, and automated diagnostic reports.

### Maintenance Recommendations

Delivers actionable maintenance procedures (e.g., "Inspect sensor calibration, run manual inspection, verify baseline telemetry").

# Implementation Approach & Prototype Flow

Sensors › ESP32 › Edge AI › MQTT › Dashboard › Maintenance Rec.

### 1. Hardware Integration

- ESP32 Dual-Core running FreeRTOS tasks.
- MPU6050 sampling tri-axial acceleration.
- Hardware relay interlock for emergency trip.

### 2. TinyML Deployment

- Spectral feature extraction via Edge Impulse.
- On-device classification model state evaluation.
- Compiled C++ library executed locally on MCU.

### 3. Application Stack

- Mosquitto MQTTS broker for messaging.
- Python FastAPI server for logging.
- React + TypeScript + Tailwind operator UI.

PROTOTYPE MATURITY STAGE

**Functional Prototype & Tested Proof-of-Concept**

Validated on motor test bench for real-time vibration classification, anomaly flagging, and automated power cutoff.

# Market Readiness & Commercialisation Plan

MARKET & ADOPTION

### Target Market & Beneficiaries

- Manufacturing plants, textile facilities, automotive assembly, and power generation units.
- MSMEs looking for high-value predictive maintenance without expensive capital expenditure.
- Plant maintenance managers seeking real-time health score metrics.

### Industry Standard Alignment

- Compliant with ISO standards for mechanical vibration evaluation on rotating machinery.
- Uses standard industrial telemetry protocols (MQTT, Modbus ready).
- Easy retrofit onto existing motor housings without structural alterations.

### Go-To-Market (GTM) Strategy

Equipment Health-as-a-Service (EHaaS) model—providing factory floors with low-cost hardware nodes coupled with monthly dashboard subscriptions.

### Industry Adoption Roadmap

Initial pilot trials in regional MSME manufacturing hubs $\to$ IP65 industrial enclosure certification $\to$ Mass domestic PCB manufacturing partnership.

# Key Technologies & Tech Stack

TECH STACK

### ESP32
Edge Computing

### MQTT
Lightweight Industrial Messaging

### Edge Impulse
TinyML Model Deployment

### Python
Backend Analytics & Ingestion

### React
Industrial Operator Interface

### Tailwind CSS & TypeScript
Responsive & Type-Safe UI Stack

**Economic Viability Model**
Low Bill of Materials (BOM) per node drastically reduces initial adoption cost for MSME factory floors.

HIGH ROI

# Future Scope & Scalability Roadmap

FUTURE ROADMAP

### Predictive Remaining Useful Life (RUL)

Advanced regression models predicting exact remaining operational hours before component replacement.

### Multi-Machine Fleet Monitoring

Centralized factory dashboard scaling across hundreds of concurrent node endpoints across production lines.

### Cloud & Hybrid Deployment

AWS/Azure IoT Hub integration for cross-plant historical fleet trend analysis.

### OTA Firmware Updates

Seamless Over-The-Air deployment of updated TinyML models directly to ESP32 nodes.

### Mobile Application

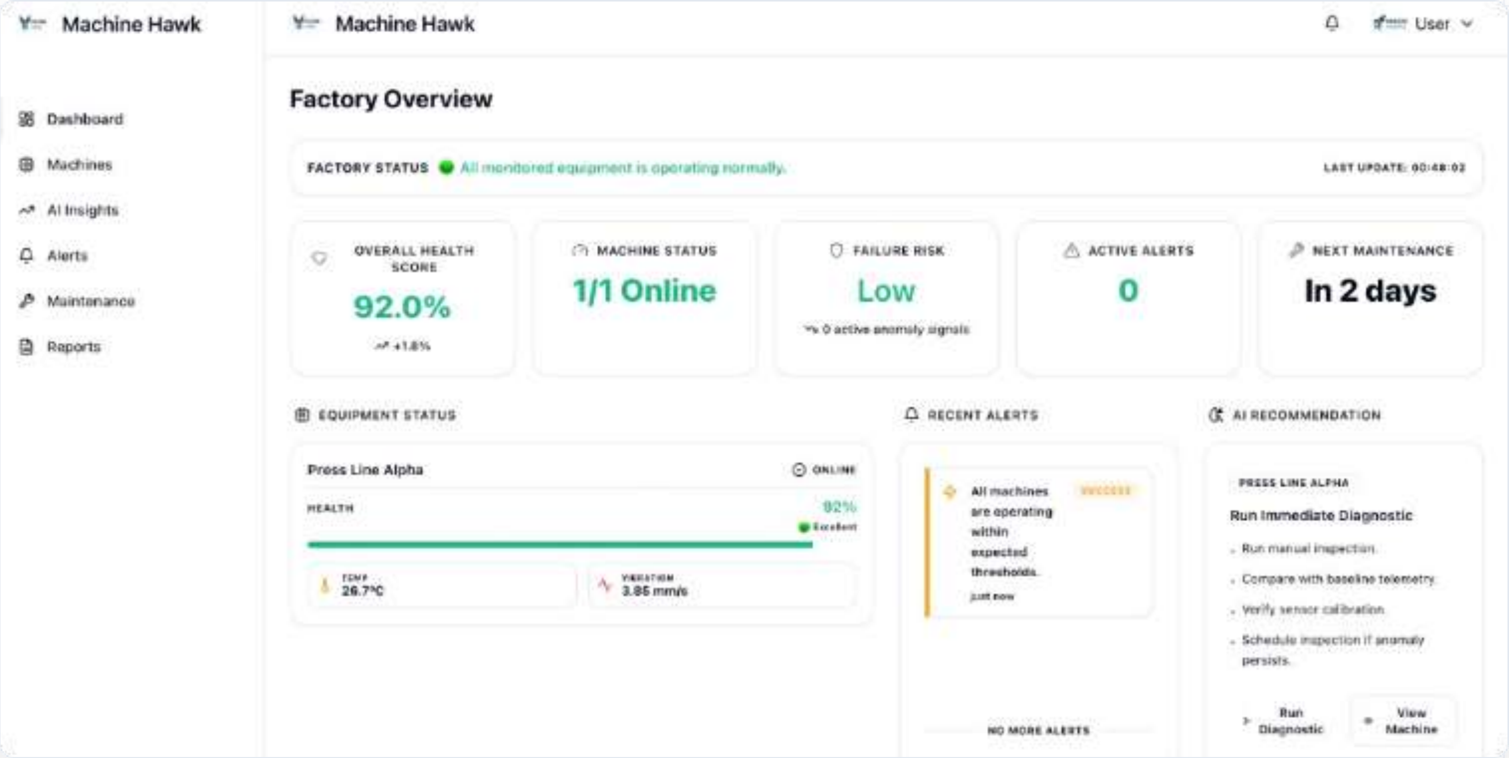
Instant push notifications and mobile diagnostic guides for floor technicians.

### Industrial Protocol Support

Native integration with Modbus, OPC-UA, and legacy industrial PLC controllers.

# Prototype Demonstration & Dashboard

PROTOTYPE DEMO




### Live Factory Overview

Real-time monitoring of machine online status, overall health scores, and risk metrics.

### Telemetry Tele-Sensors

Continuous monitoring of operating temperature (°C) and vibration acceleration (mm/s).

### AI Recommendation Engine

Automated diagnostic actions: manual inspection prompts, baseline telemetry comparison, and sensor calibration.

# Project Submission Summary

FINAL SUBMISSION

## Machine Hawk

**From Reactive Maintenance to Predictive Intelligence**

Empowering modern factories with low-cost, local edge intelligence to prevent costly equipment downtime and protect critical industrial assets.


**References & Technical Standards**

- Edge Impulse TinyML Framework & C++ SDK
- ISO Standards for Industrial Mechanical Vibration Evaluation
- OASIS MQTT v5.0 Messaging Specifications


## Thank You for Your Consideration!

Team Machine Hawk · Dr H N National College of Engineering

Ready for Jury Q&A & Live Demonstration